Homework 2: A Forecast for VELTRAXA

Relapsed/refractory mantle cell lymphoma

Applied Data Science and AI in Pharma Commercial · Columbia University · Fall 2026

Released Monday, October 19, 2026 · Due Monday, November 2, 2026, 11:59 PM ET · 10% of the course grade · Individual

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1. The brief

VELTRAXA is a fictional once-daily oral BTK degrader launching in July 2027 for relapsed/refractory mantle cell lymphoma after at least one prior line (2L+). The brand forecaster needs:

- a monthly budget for July 2027 to December 2028;
- a long-range plan to 2035, with a low, base and high case;
- an answer to one question: the plan of record says $180M in 2029. Is that reachable, and what would have to be true?

Use the model from class. Grading rewards clear, well-supported assumptions, not model complexity. All data are synthetic.

2. Data (data/homework2/)

File	Use
mcl_epi_inputs.csv	New MCL cases per year (sum the age bands)
mcl_treatment_funnel.csv	Diagnosed, treated and progression rates, with ranges
mcl_market_monthly.csv	Observed 2L and 3L+ starts, 2019–2026, to check your funnel
mcl_market_by_region.csv	Regional shares for the top-down split
mcl_analog_launches_monthly.csv, _attributes.csv	Six analog launches: peak share and uptake by year
veltraxa_assumptions.csv	Starter assumption book: launch, price, gross-to-net, compliance, patent expiry

You may start from Week_06_LYMBREXA_Forecast_Model.xlsx or the class notebook. Change any assumption, but record why. Do not change the data.

3. Tasks

1. **Assumption book (30 pts).** At least 20 rows in the class format (ID, category, assumption, unit, low, base, high, source type, source, reason). Every number in the model traces to a row. Support the key rows with the data: progression rates from the funnel file, peak share and uptake from the analogs and the order-of-entry rule, months on therapy from the label and analogs.
2. **Funnel and brand forecast (25 pts).** Annual, 2023–2035: new cases, 1L, 2L and 3L+ starts, new VELTRAXA patients, patients on therapy, net revenue. State the model type (incidence or prevalence; flow or stock). Show that the funnel matches the 2025 market within about 10%, or explain the gap.
3. **Ranges (15 pts).** Low, base and high scenarios, and a tornado chart for 2029 revenue. Name the three assumptions that matter most.

4. **Checks (10 pts).** Compare a simple trend of 2L+ starts with your funnel for 2027–28. Split 2028 new patients by region so the regions sum to the national total.
5. **Memo (20 pts).** Two pages to the brand forecaster: the forecast and its range in one chart; the three key assumptions and the evidence that would change each; your answer on $180M; what you would research in the next 90 days. Include the monthly budget and the annual long-range plan from the same model, with the phasing assumption stated.

Note on duration. VELTRAXA is taken until progression, so many patients stay on it for more than a year. Booking all of a patient's revenue in the start year is acceptable; record it as an assumption.

Bonus (+5). Add a foundation-model forecast of 2L+ starts (e.g., Chronos-Bolt) next to your trend check, with one paragraph comparing the two.

4. Submission

Deliverable	File name
Model: Excel workbook or executed notebook	HW2_<uni>_model.xlsx or HW2_<uni>.ipynb
Assumption book	HW2_<uni>_assumptions.csv
Memo (PDF, 2 pages)	HW2_<uni>_memo.pdf

Submit on the course site. Late work, collaboration and AI use follow the syllabus policies. Questions posted on the course forum by Friday, October 30 are answered before the deadline.

5. Rubric

Component	Points	Full credit	Common deductions
Assumption book	30	20+ rows; every number sourced or reasoned; key rows supported by data	Unsourced numbers; unexplained ranges; model inputs missing from the book
Funnel and forecast	25	Correct chain with a one-year lag; model type stated; 2025 check	Share applied to all patients instead of line starts; no lag; list price used as net
Ranges	15	Scenarios and tornado; top three assumptions named	A single number
Checks	10	Trend compared with funnel; regions sum to national	Regions that do not add up
Memo	20	Answer and range first; three assumptions; clear $180M verdict; 90-day plan	Describes the steps instead of answering; no range